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Systems and methods for wavelength locking in optical sensing systems

Abstract

Disclosed herein is an integrated photonics device including a frequency stabilization subsystem for monitoring and/or adjusting the wavelength of light emitted by one or more light sources. The device can include one or more selectors that can combine, select, and/or filter light along one or more light paths, which can include light emitted by a plurality of light sources. Example selectors may include, but are not limited to, an arrayed waveguide grating (AWG), a ring resonator, a plurality of distributed Bragg reflectors (DBRs), a plurality of filters, and the like. Output light paths from the selector(s) can be input into one or more detector(s). The detector(s) can receive the light along the light paths and can generate one or more signals as output signal(s) from the frequency stabilization subsystem. A controller can monitor the wavelength and can adjust or generate control signal(s) for the one or more light sources to lock the monitored wavelength to a target wavelength (or within a targeted range of wavelengths).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/373,167, filed Jul. 12, 2021, which is a continuation of U.S. patent application Ser. No. 16/582,949, filed Sep. 25, 2019, now U.S. Pat. No. 11,064,592, which is a nonprovisional of, and claims the benefit under 35 USC 119(e) of, U.S. Patent Application No. 62/738,649, filed Sep. 28, 2018, the contents of which are incorporated herein by reference as if fully disclosed herein.

FIELD

(1) This disclosure relates generally to optical sensing systems. More particularly, this disclosure relates to methods for wavelength locking and associated optical sensing systems.

BACKGROUND

(2) Optical sensing systems can be useful for many applications. In some instances, it may be useful to measure the optical properties of light emitted by light sources included in the optical sensing systems. For example, the optical properties of emitted light can be monitored to ensure that a light source is tuned to a target wavelength and/or has a certain amount of wavelength stability.

(3) In some examples, the light sources can emit light having different properties such as different wavelengths. The emitted light can propagate along one or more light paths. The optical sensing system can also include one or more multiplexers for combining the emitted light along the light paths, where the multiplexer(s) can be component(s) separate from those measuring the optical properties of the emitted light. Separate components for the two functions may increase the size, cost, and complexity of the optical sensing system.

SUMMARY

(4) Disclosed herein is an integrated photonics device including a frequency stabilization subsystem for monitoring and/or adjusting the wavelength of light emitted by one or more light sources. The device can include one or more selectors which can combine, select, and/or filter light along one or more light paths, wherein the light can include light emitted by a plurality of light sources. Selectors may include, but are not limited to, an arrayed waveguide grating (AWG), a ring resonator, a plurality of distributed Bragg reflectors (DBRs), a plurality of filters, and the like. Output light paths from the selector(s) can be input into one or more detector(s). The detector(s) can receive the light along the light paths and can generate one or more signals as output signal(s) from the frequency stabilization subsystem. A controller can monitor the wavelength and can adjust or generate control signal(s) for the one or more light sources to lock the monitored wavelength to a target wavelength (or within a targeted range of wavelengths).

(5) A system is disclosed which may include at least a plurality of light sources that may emit a light in response to one or more control signals, where the light may propagate along one or more first light paths, a frequency stabilization subsystem that may receive at least a portion of the one or more first light paths, where the frequency stabilization subsystem may include one or more selectors that combine, select, filter, or a combination thereof, of at least a portion of the one or more first light paths, and one or more detectors that may receive light along one or more second light paths from the one or more selectors and may generate one or more signals. In some examples, the one or more signals may be one or more outputs from the frequency stabilization subsystem. The frequency stabilization subsystem may also include a controller that may receive the one or more signals from the frequency stabilization subsystem, may determine a monitored wavelength from the one or more signals, may determine a difference between the monitored wavelength and a targeted wavelength, and may generate the one or more control signals based on the difference.

(6) Additionally or alternatively, in some examples, the one or more selectors may include an arrayed waveguide grating (AWG), the AWG including a plurality of waveguides, where at least two of the plurality of waveguides may have different lengths. The AWG may output the light

along the one or more second light paths having different phase shifts. Additionally or alternatively, in some examples, the frequency stabilization subsystem may include a plurality of attenuators coupled to the AWG and the one or more detectors, the plurality of attenuators modulating the light along the one or more second light paths at different frequencies. Additionally or alternatively, in some examples, the one or more selectors may include a first ring resonator and a second ring resonator, the first ring resonator including a first looped waveguide which may optically couple at least a portion of the one or more first light paths having a first set of wavelengths, and the second ring resonator including a second looped waveguide which may optically couple at least a portion of the one or more first light paths having a second set of wavelengths.

(7) Additionally or alternatively, in some examples, the one or more selectors may include a distributed Bragg reflector (DBR) component, the DBR component including a first DBR having a first structure that outputs at least a portion of the one or more first light paths having a first set of wavelengths, and a second DBR that outputs the at least the portion of the one or more first light paths having a second structure that outputs a second set of wavelengths. Additionally or alternatively, in some examples, the one or more selectors may include a filter component, the filter component including a waveguide, a first filter that filters at least a portion of the one or more first light paths having a first set of wavelengths, and a second filter that filters at least a portion of the one or more first light paths having a second set of wavelengths. Additionally or alternatively, in some examples, the one or more selectors may include one or more of a resonator, a multiplexer, an echelle grating multiplexer, a Mach-Zehnder interferometer, a Fabry-Perot cavity, a nanobeam cavity, and so forth. Additionally or alternatively, in some examples, the one or more detectors may include a single detector, and the one or more selectors may be coupled to different sides of the single detector.

(8) Additionally or alternatively, in some examples, the one or more selectors may include one or more of an arrayed waveguide grating (AWG), a resonator, a distributed Bragg reflector (DBR) component, a filter component, a multiplexer, an echelle grating multiplexer, a Mach-Zehnder interferometer, a Fabry-Perot cavity, a nanobeam cavity and so forth. Additionally or alternatively, in some examples, the one or more detectors may include a twin waveguide detector having two waveguides on a same substrate, where a spacing between the two waveguides may be less than a longitudinal dimension of a given waveguide. Additionally or alternatively, in some examples, the one or more selectors may include one or more sensors measuring a temperature of the one or more selectors and one or more heaters for adjusting the temperature of the one or more selectors.

Additionally or alternatively, in some examples, the frequency stabilization subsystem may further include a multiplexer, the multiplexer receiving the one or more first light paths and outputting a third light path, the third light path being the at least the portion of the one or more first light paths, where the one or more selectors select from the third light path and output a fourth light path, the fourth light path included in the one or more outputs of the frequency stabilization subsystem.

Additionally or alternatively, in some examples, the one or more selectors may select from the one or more first light paths and outputs a fourth light path, the fourth light path included in the one or more outputs of the frequency stabilization subsystem. Additionally or alternatively, in some examples, the frequency stabilization subsystem may further include a multiplexer, the multiplexer receiving the one or more first light paths and outputting a third light path, the frequency stabilization subsystem further including a splitter that receives the third light path and outputs a fourth light path and a fifth light path, the fourth light path included in the one or more outputs of the frequency stabilization subsystem, and the fifth light path being the at least the portion of the one or more first light paths.

(9) A method for operating one or more light sources in an optical sensing system is disclosed. The method can include sending one or more control signals to one or more light sources, emitting light from the one or more light sources, where the emitted light is based on the one or more control signals, propagating at least a portion of the emitted light along one or more first light paths to a

frequency stabilization subsystem, performing one or more of combining, multiplexing, filtering, and selecting at least a portion of the one or more first light paths using one or more selectors included in the frequency stabilization subsystem, receiving light along one or more second light paths by one or more detectors, generating one or more signals using the one or more detectors, the one or more signals indicative of the light along the one or more second light paths, determining a monitored wavelength from the one or more signals, determining a difference between the monitored wavelength and a target wavelength, and adjusting the one or more control signals to the one or more light sources based on the difference.

(10) Additionally or alternatively, in some examples, the performance of one or more of combining, multiplexing, filtering, and selecting the at least the portion of the one or more first light paths may include creating a first phase shift in the at least the portion of the one or more first light paths having a first set of wavelengths using an arrayed waveguide grating (AWG) and creating a second phase shift in the at least the portion of the one or more first light paths having a second set of wavelengths using the AWG. Additionally or alternatively, in some examples, the performance of one or more of combining, multiplexing, filtering, and selecting the at least the portion of the one or more first light paths may include optically coupling at least the portion of the one or more first light paths having a first set of wavelengths using a first ring resonator and optically coupling the at least the portion of the one or more first light paths having a second set of wavelengths using a second ring resonator. Additionally or alternatively, in some examples, the method may further include determining the target wavelength by taking a ratio of the one or more signals and determining an overlapping wavelength of the one or more signals. Additionally or alternatively, in some examples, the method may further include measuring a temperature of the one or more selectors using one or more sensors, determining whether the temperature meets one or more temperature criteria, and in accordance with the temperature not meeting the one or more temperature criteria, adjusting the temperature using one or more heaters.

(11) A method for forming an optical sensing system is disclosed. The method can include providing a plurality of light sources, the plurality of light sources emitting light that propagates along one or more first light paths, providing a controller, arranging the plurality of light sources to receive one or more control signals from the controller, forming a frequency stabilization subsystem and arranging the frequency stabilization subsystem to receive the one or more first light paths. In some examples, forming the frequency stabilization subsystem may include forming one or more selectors to combine, select, filter, or a combination thereof at least a portion of the one or more first light paths, forming and arranging one or more detectors to receive light along one or more second light paths from the one or more selectors, arranging the one or more detectors to generate one or more outputs from the frequency stabilization subsystems, and arranging and connecting the controller to the frequency stabilization subsystem and the plurality of light sources. Additionally or alternatively, in some examples, forming the one or more selectors may include forming one or more of an arrayed waveguide grating (AWG), a ring resonator, a distributed Bragg reflector (DBR) component, a filter component, a multiplexer, an echelle grating multiplexer, a Mach-Zehnder interferometer, a Fabry-Perot cavity, a nanobeam cavity, and so forth.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like structural elements, and in which:

(2) FIG. 1A illustrates a block diagram of a portion of an example integrated photonics device including a frequency stabilization subsystem, which can include a selector that selects and outputs

a light path;

(3) FIG. 1B illustrates an example plot of multiple signals and a target wavelength;

(4) FIG. 2 illustrates a block diagram of a portion of an example integrated photonics device including a frequency stabilization subsystem, which can include a selector that combines multiple input light paths and outputs a selected light path;

(5) FIG. 3 illustrates a block diagram of a portion of an example integrated photonics device including a frequency stabilization subsystem, which can include a selector that receives light paths from a splitter;

(6) FIG. 4 illustrates a block diagram of an example device including multiple frequency stabilization subsystems;

(7) FIGS. 5A-5B illustrate block diagrams of example frequency stabilization subsystems including an arrayed waveguide grating and multiple detectors;

(8) FIG. 6 illustrates a block diagram of an example frequency stabilization subsystem including ring resonators and multiple detectors;

(9) FIGS. 7A-7B illustrate block diagrams of example frequency stabilization subsystems including a distributed Bragg reflector and multiple detectors;

(10) FIGS. 8A-8B illustrate block diagrams of example frequency stabilization subsystems including a filter component and multiple detectors;

(11) FIG. 9 illustrates a flow chart of an example operation of an optical sensing system having multiple detectors;

(12) FIG. 10A illustrates a block diagram of an example frequency stabilization subsystem including an arrayed waveguide grating and a common detector;

(13) FIG. 10B illustrates a block diagram of an example frequency stabilization subsystem including a ring resonator and a common detector;

(14) FIG. 10C illustrates a block diagram of an example frequency stabilization subsystem including a distributed Bragg reflector component and a common detector;

(15) FIG. 11 illustrates a cross-sectional view of an example detector including multiple waveguides formed on the same substrate;

(16) FIG. 12 illustrates a block diagram of an example optical sensing system including a frequency stabilization subsystem including an arrayed waveguide grating and a detector;

(17) FIG. 13A illustrates a block diagram of an example ring resonator and calibration components; and

(18) FIG. 13B illustrates a block diagram of an example distributed Bragg reflector and calibration components.

(19) The use of cross-hatching or shading in the accompanying figures is generally provided to clarify the boundaries between adjacent elements and also to facilitate legibility of the figures.

Accordingly, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, element proportions, element dimensions, commonalities of similarly illustrated elements, or any other characteristic, attribute, or property for any element illustrated in the accompanying figures.

(20) Additionally, it should be understood that the proportions and dimensions (either relative or absolute) of the various features and elements (and collections and groupings thereof) and the boundaries, separations, and positional relationships presented between them, are provided in the accompanying figures merely to facilitate an understanding of the various embodiments described herein and, accordingly, may not necessarily be presented or illustrated to scale, and are not intended to indicate any preference or requirement for an illustrated embodiment to the exclusion of embodiments described with reference thereto.

DETAILED DESCRIPTION

(21) In the following description of examples, reference is made to the accompanying drawings in which are shown, by way of illustration, specific examples that can be practiced. It is to be

understood that other examples can be used and structural changes can be made without departing from the scope of the various examples.

(22) Various techniques and process flow steps will be described in detail with reference to examples as illustrated in the accompanying drawings. In the following description, numerous specific details are set forth in order to provide a thorough understanding of one or more aspects and/or features described or referenced herein. It will be apparent, however, to one skilled in the art, that one or more aspects and/or features described or referenced herein may be practiced without some or all of these specific details. In other instances, well-known process steps and/or structures have not been described in detail in order not to obscure some of the aspects and/or features described or referenced herein.

(23) Further, although process steps or method steps can be described in a sequential order, such processes and methods can be configured to work in any suitable order. In other words, a sequence or order of steps described in this disclosure does not, in and of itself, indicate a requirement that the steps be performed in that sequence or order. Further, some steps may be performed simultaneously despite being described or implied as occurring non-simultaneously (e.g., because one step is described after the other step). Moreover, the illustration of a process by its description in a drawing does not imply that the illustrated process is exclusive of other variations and modification thereto, does not imply that the illustrated process or any of its steps are necessary to one or more of the examples, and does not imply that the illustrated process is preferred.

(24) Various reference characters are used throughout the description for purposes of referring to one or more elements drawn in the figures. In some instances, a reference character may include numbers followed by a letter. Other reference characters may include the same numbers, but followed by a different letter. The description may refer to the group of elements by referring to the common numbers only, where no letter is appended. In such case, the reference to the group of elements means that the disclosed examples apply to one or more of the group of elements.

(25) Disclosed herein is an integrated photonics device including a frequency stabilization subsystem for monitoring and/or adjusting the wavelength of light emitted by one or more light sources. The frequency stabilization subsystem can include selector(s) which can combine, select, and/or filter light along one or more light paths, which can include light emitted by a plurality of light sources. Example selectors may include, but are not limited to, an arrayed waveguide grating (AWG), a plurality of ring resonators, a plurality of distributed Bragg reflectors (DBRs), a plurality of filters, and the like. Output light paths from the selector(s) can be input into one or more detector(s). The detector(s) can receive the light along the light paths and can generate one or more signals as output signal(s) from the frequency stabilization subsystem. A controller can monitor the wavelength and can adjust or generate control signal(s) for the one or more light sources to lock the monitored wavelength to a target wavelength (or within a targeted range of wavelengths).

(26) Representative applications of methods and apparatus according to the present disclosure are described in this section. These examples are being provided solely to add context and aid in the understanding of the described examples. It will thus be apparent to one skilled in the art that the described examples may be practiced without some or all of the specific details. Other applications are possible, such that the following examples should not be taken as limiting.

(27) Example Frequency Stabilization Subsystems

(28) In some examples, monitoring the optical properties of emitted light may employ the use of a frequency stabilization subsystem. The frequency stabilization subsystem can be used to verify whether the light sources are tuned to a target wavelength and/or have a certain amount of wavelength stability and, in some examples, make adjustments accordingly. The frequency stabilization subsystem **108** can be configured with one or more functions: multiplexing, combining, selecting, filtering, etc. In some examples, the number of components in the device **100** may include fewer or more components.

(29) In some examples, the frequency stabilization subsystem can include a selector that selects and

outputs light on a selected light path. FIG. 1A illustrates a block diagram of a portion of an example integrated photonics device including a frequency stabilization subsystem, which can include a selector that selects and outputs light on a light path. In FIG. 1A, an optical sensing system can include a device **100** which may include one or more light sources **102** and a multiplexer **146** which may be part of a frequency stabilization subsystem **108**. The light sources **102** can be configured to emit light along one or more light paths **144**. In some examples, the light sources **102** can be configured to emit light having different ranges of wavelengths.

(30) The light paths **144** can be input into the frequency stabilization subsystem **108**. The frequency stabilization subsystem **108** may include a multiplexer **146**, a selector **110**, and one or more detectors **131**. The multiplexer **146** can receive the emitted light along the light path(s) **144** and can combine the emitted light to output light path **148**. The light path **148** can be input to the selector **110**. The selector **110** can select and output the light on light path **142**, which can be one of the outputs of the frequency stabilization subsystem **108**. The selector **110** can also generate outputs along the light paths **140** to the detectors **131A** and **131B**. The detectors **131A** and **131B** can receive the light paths **140** and output a signal or signals to a controller **112**. Even though the detectors **131A** and **131B** illustrate a combined, single output to the controller **112**, in some examples, an individual output from each of the detectors **131A** and **131B** may be output to the controller **112** such that the controller receives an output from detector **131A** and an output from detector **131B**.

(31) The selector **110** can be a component that receives light via an input light path, selects light from the input light path, and outputs the selected light. In some examples, the selector **110** can output multiple light paths, which can include at least two different wavelengths (discussed below). For example, the selector **110** can receive light on the light path **148** as input and can output light along light paths **142** and **140**. The selector **110** can include one or more passive components such as a filter, resonator, multiplexer, an echelle grating multiplexer, an arrayed waveguide grating (AWG), Mach-Zehnder interferometer (MZI), a Fabry-Perot cavity, a nanobeam cavity, a ring resonator, a Distributed Bragg Reflector (DBR), or the like for combining, selecting, and/or filtering light. The terms “frequency stabilization” and “frequency stabilization subsystem” as used throughout this disclosure can refer to frequency stabilization, wavelength tuning, or both.

(32) The detectors **131** can include any type of diode that can respond to or measure photons impinging on its active area. The detectors **131** can generate one or more signals indicative of the light along the light paths **140**; these one or more signals can be an output signal or output signals from the frequency stabilization subsystem **108** to the controller **112**, for example.

(33) Although FIG. 1A illustrates the multiplexer **146** as included in the frequency stabilization subsystem **108**, examples may include the multiplexer **146** as being a component separate from the frequency stabilization subsystem **108** (not shown in FIG. 1A). Examples of the device **100** can further include one or more additional components, such as filters, amplifiers, analog-to-digital converters (ADCs), etc. (not shown) located between the detector(s) and the controller **112**. These additional components can perform one or more operations on or with the signals from the detector(s) such as processing the signals, amplifying the signals, performing one or more calculations or comparisons, and so forth.

(34) The signal(s) from the frequency stabilization subsystem **108** can be used as feedback in, for example, a control loop. The device **100** can also include a controller **112** that can receive and analyze the signal(s) from the frequency stabilization subsystem **108**. The controller **112** can generate one or more signals that can be inputs to the light sources **102**. In some examples, the analysis can include monitoring the wavelength of the emitted light as discussed herein and determining the difference between the monitored wavelength and a target wavelength. The controller **112** may be configured to lock the monitored wavelength to the target wavelength.

(35) The signal(s) from the frequency stabilization subsystem **108** can be the same signals used to control the light sources **102** (e.g., control signal(s)) and the properties of light emitted by the light

sources **102** along the light paths **144**. In some examples, the signal(s) from the frequency stabilization subsystem **108** can be indicative of changes in one or more properties (e.g., temperature, current, etc.) of the light sources **102**. The changes may be associated with locking the monitored wavelength to the target wavelength. In some examples, the controller **112** can use other information (e.g., measured temperature of the light sources **102**) in generating the signal(s).

(36) In some examples, the frequency stabilization subsystem **108** outputs multiple signals to the controller **112**. In such instances, the controller **112** can lock the wavelength at or near the crossing points (e.g., overlap) of the signals, as shown in FIG. **1B**. In some examples, the wavelength locking may occur near the crossing points so long as signals **133A** and **B** exceed the target signal to noise ratio. Although not illustrated in FIG. **1B**, in some examples, the wavelength outputs may generally approximate a Gaussian curve. The selector **110** can select light on light paths (from the input light path **148**) that include sets of wavelengths. In some examples, at least two light paths can include one or more different wavelengths and one or more common wavelengths. A target wavelength **135** can be included in the common wavelength(s), and the selector **110** can select the light on the light paths, such that the range of wavelengths is close to the target wavelength **135**. For example, the selector **110** can select a first light path associated with signal **133A** and a second light path associated with signal **133B**. Signal **133A** can be the signal generated by the detector **131A** (in FIG. **1A**), and signal **133B** can be the signal generated by the detector **131B** (in FIG. **1A**). The first set of wavelengths may include wavelengths shorter than the target wavelength **135**, and the second set of wavelengths may include wavelengths longer than the target wavelength **135**.

(37) The signals may have an intensity that varies with wavelength, and a maximum intensity (e.g., maximum current value) located at one of the wavelengths. The difference between the wavelength of the maximum intensity and the target wavelength **135** can be referred to as a wavelength spacing **137**. Although FIG. **1B** shows the wavelength spacings **137** for both signals **133A** and **133B** as being the same, examples may include different wavelength spacings **137**. In instances where the wavelength spacings **137** may differ from one another, the ratio of the signals may affect the location of the target wavelength **135**.

(38) The controller **112** can receive the associated signals **133** and can lock the monitored wavelength to the target wavelength **135**, and the target ratio can be determined by taking the ratio of the signals **133A** and **133B**. If the monitored wavelength is not within a certain threshold wavelength from the target wavelength **135**, the controller **112** can adjust or send a new signal to the light sources **102** (or another controller that controls the light sources **102**).

(39) In some instances, the light sources **102** may emit light from at least two of the light sources at different times. For example, the light sources **102** can be activated sequentially, one at a time. The frequency stabilization subsystem **108** can monitor the wavelength of the emitted light, and the controller **112** can adjust the individual signals to one or more light sources. Alternatively, the controller **112** can receive signals from the detectors sequentially, and the controller can adjust the signals to the light sources in response to the plurality of sequentially received signals from the detectors.

(40) Although FIG. **1B** illustrates the target wavelength **135** as being a single wavelength, examples may include a target range of wavelengths. In some instances, the target wavelength **135** may change during device operation, and the device **100** can be configured for locking to different target wavelengths at different times. Additionally or alternatively, the device can lock the monitored wavelength to a target ratio (which can be the ratio of the signals from the detectors). The target ratio may, in some examples, be different at different times.

(41) In some examples, the frequency stabilization subsystem can include a selector that combines the input light on light paths along with selecting and outputting light on a selected light path. FIG. **2** illustrates a block diagram of a portion of an example integrated photonics device including a frequency stabilization subsystem, which can include a selector that combines light on multiple input light paths and outputs light on a selected light path. The device **200** can include one or more

light sources **202** and a frequency stabilization subsystem **208**. The light sources **202** and its output light paths **244** can be correspondingly similar in functionality and structure as the light sources **102** and the light paths **144** of FIG. 1A, respectively.

(42) Light on the light paths **244** can be input into the frequency stabilization subsystem **208**. The frequency stabilization subsystem **208** can include a selector **210** and one or more detectors **231**. The detectors **231** can be correspondingly similar in functionality and structure as the detectors **131** of FIG. 1A.

(43) The selector **210** can be configured with multiple functions, such as the functions of the multiplexer **146** and the selector **110** of FIG. 1A. The selector **210** can receive the emitted light along the light paths **244** and can combine the emitted light. The selector **210** can select from the light combined on the emitted light path and output the selected light on light path **242**, which can be one of the outputs of the frequency stabilization subsystem **208**. The selector **210** can also output light along the light paths **240** to the detectors **231A** and **231B**, and the detectors **231A** and **231B** can output signals to a controller **212**. The controller **212** can be correspondingly similar in functionality and structure as controller **112** of FIG. 1A. Although the output signals of detectors **231A** and **231B** are illustrated in FIG. 2 as being combined before being received by controller **212**, in some examples, the output signals of detectors **231A** and **231B** may be individual output signals separately received by the controller **212**.

(44) In some examples, the frequency stabilization subsystem can include a selector that receives light paths from a splitter. FIG. 3 illustrates a block diagram of a portion of an example integrated photonics device including a frequency stabilization subsystem, which can include a selector that receives light paths from a splitter. The device **300** can include one or more light sources **302** and a frequency stabilization subsystem **308**. The light sources **302** and its output light paths **344** can be correspondingly similar in functionality and structure as the light sources **102** and **144** of FIG. 1A, respectively.

(45) The light paths **344** can be input into the frequency stabilization subsystem **308**. The frequency stabilization subsystem **308** can include a multiplexer **346**, a splitter **355**, a selector **310**, and one or more detectors **331**. The multiplexer **346** and the detectors **331** can be correspondingly similar in functionality and structure as the multiplexer **146** and detectors **131** of FIG. 1A, respectively.

(46) The multiplexer **346** can receive the emitted light along the light paths **344** and can combine the emitted light on light path **348**. The light path **348** can be input to the splitter **355**. The splitter can split the light on light path **348** to output light on light path **342** and light path **338**. The light path **342** can be one of the outputs of the frequency stabilization subsystem **308**.

(47) The light path **338** can be input to the selector **310**. The selector **310** can generate outputs of light along the light paths **340** to the detectors **331A** and **331B**. The detectors **331** can receive the light paths **340** and can output signals to a controller **312**. The controller **312** can be correspondingly similar in functionality and structure as controller **112** of FIG. 1A.

(48) The splitter **355** can split the light along the light path **348** such that light along the light path **348** has a different proportion of light intensity than the light along the light path **338**. In some examples, the split can be such that light along the light path **338** includes a lower proportion (of light from the light path **348**) than light along the light path **342**. For example, the light path **338** can include less than 10% of light from the light path **348**, and the light path **342** can include more than 90% of light from the light path **348**.

(49) Although FIG. 3 illustrates the multiplexer **346** as included in the frequency stabilization subsystem **308**, examples of the disclosure can include the multiplexer **346** as being a component separate from the frequency stabilization subsystem **308**. Examples of the disclosure can further include one or more additional components, such as filters, amplifiers, analog-to-digital converters (ADCs), etc. (not shown) located between the detector(s) and the controller **312**. These additional components can perform one or more operations on or with the signals from the detector(s) such as processing the signals, amplifying the signals, performing one or more calculations or comparisons,

or any combination thereof, and so forth.

(50) In some examples, the device can include multiple frequency stabilization subsystems for monitoring the wavelength of light emitted by multiple light sources, as shown in FIG. 4. Device **400** can include light sources **402**, a plurality of frequency stabilization subsystems **408**, and a controller **412** that are correspondingly similar in functionality and structure as device **100** (or device **200** or **300**), light sources **102** (or light sources **202** or **302**), frequency stabilization subsystems **108** (or frequency stabilization subsystems **208** or **308**), and controller **112** (or controller **212** or **312**), respectively of FIGS. 1A, 2, and/or 3. Similarly, light paths **444** and light path **442** can be correspondingly similar in functionality and structure as the light paths **144** (or light paths **244** or **344**) of FIGS. 1A, 2, and/or 3. Although the frequency stabilization subsystems may be illustrated in FIG. 4 as separate elements which are not communicating, in some examples, the frequency stabilization subsystems may be configured to communicate with one another as well as with other elements of device **400**.

(51) Selector

(52) Examples of the disclosure can include a selector that includes an AWG. FIG. 5A illustrates a block diagram of a portion of an example frequency stabilization subsystem including an AWG and multiple detectors. Frequency stabilization subsystem **508A** can receive one or more light paths **541** as inputs and can generate one or more outputs, such as output **549A**, and output **549B**. In some examples, the frequency stabilization subsystem **508A** can include AWG **510A** and detectors **531A** and **531B**. The detectors **531** can be correspondingly similar in function and structure as detectors **131** of FIG. 1A.

(53) The AWG **510A** can include a plurality of waveguides **515A**, where at least two waveguides can have different lengths. The phase shift induced by the different waveguides **515A** of different lengths can vary. The outputs of the AWG **510A** can have different phase shifts, which may be wavelength dependent. The light path **541** can include light having a plurality of wavelengths and can be multiplexed using the AWG **510A**. One output from the AWG **510A**, such as the one received by the detector **531A**, may include a first set of wavelengths, and another output from the AWG **510B**, such as the one received by the detector **531B**, may include a second set of wavelengths. For example, the output **549A** can include a first set of wavelengths, and the output **549B** can include a second set of wavelengths, where the first and second sets can include wavelengths that are different from the other set in addition to wavelengths that are common. In some examples, the sets of wavelengths can have a non-zero wavelength spacing (e.g., wavelength spacing **137** illustrated in FIG. 1B) from the target wavelength (e.g., target wavelength **135** illustrated in FIG. 1B).

(54) The detector **531A** and the detector **531B** may receive light from the AWG **510A** and may generate signals indicative of the measured light. The signals may be transmitted along the output **549A** and the output **549B**. In some examples, the signals may be current signals. A controller (e.g., controller **112** illustrated in FIG. 1A) may receive the output signals, may take the ratio of the signals, and may lock the monitored wavelength to a targeted wavelength by transmitting a signal to the light sources which may provide an adjustment to the light sources, for example, light sources **102** of FIG. 1A.

(55) In some instances, the AWG may also output light along a light path, as shown in FIG. 5B. Frequency stabilization subsystem **508B** may include an AWG **510B**, which may be a selector that is configured to output light on a selected light path such as the light path **542**. The light path **542** may be an output of the frequency stabilization subsystem **508B**. The AWG **510B** can receive light on multiple light paths as inputs. For example, the light paths **144** of FIG. 1A or the light paths **244** of FIG. 2 may be connected to the light paths **541**.

(56) As discussed herein, the selector in the frequency stabilization subsystem can include a ring resonator. FIG. 6 illustrates a block diagram of a portion of an example frequency stabilization subsystem including one or more ring resonators and multiple detectors.

(57) Frequency stabilization subsystem **608** can receive one or more light paths **641** as inputs and can generate one or more outputs, such as output **649A** and output **649B**. The frequency stabilization subsystem **608** can include a first ring resonator **610A**, a second ring resonator **610B**, and a plurality of detectors **631A** and **631B**. The detectors **631** can be correspondingly similar in function and structure as detectors **131** of FIG. **1A**.

(58) The ring resonators **610** can include waveguides. At least one waveguide can be a looped optical waveguide. At least one waveguide can be an input waveguide that receives the light path **641** as input, and two or more waveguides can be output waveguides. In some examples, the input and output waveguides may include the same structure.

(59) Light from the light path **641** can travel through the input waveguide, and light can optically couple to a given output waveguide depending on its respective resonant mode. The first ring resonator **610A** can optically couple light having a first set of wavelengths and can output the light to the detector **631A**. The second ring resonator **610B** can optically couple light having a second set of wavelengths and can output the light to the detector **631B**. In this manner, both the first ring resonator **610A** and the second ring resonator **610B** can filter certain wavelengths of light which may pass through the ring resonators from the light path **641**.

(60) The detector **631A** and the detector **631B** may receive light from the first ring resonator **610A** and the second ring resonator **610B**, respectively, and may generate signals indicative of the monitored light. The signals may be transmitted along output **649A** and output **649B**. In some examples, the signals may be current signals. A controller, for example, the controller **112** illustrated in FIG. **1A**, may receive the signals, may take the ratio of the signals, and may lock the monitored wavelength to a targeted wavelength by transmitting a signal to the light sources which may provide an adjustment to the light sources, for example, light sources **102** of FIG. **1A**. In some examples, multiple sets of ring resonators may be employed in the frequency stabilization subsystem. Different sets of ring resonators may have waveguides with different properties, such that different sets can be used at different times.

(61) The selector in the frequency stabilization subsystem may also include a DBR component. FIG. **7A** illustrates a block diagram of a portion of an example frequency stabilization subsystem including a DBR component and multiple detectors. Frequency stabilization subsystem **708A** can receive light on one or more light paths **741** as input and can generate one or more outputs, such as output **749A** and output **749B**. The frequency stabilization subsystem **708A** can include a DBR component **710A** and the detectors **731A** and **731B**. The detectors **731** can be correspondingly similar in function and structure as detectors **131** of FIG. **1A**.

(62) In some examples, the DBR component **710A** can include a splitter **711** and a plurality of DBRs **713**. The splitter **711** can split the light received along the light path **741** into multiple outputs that can be received by the DBRs **713**. An example splitter can include, but is not limited to, a Y-junction splitter, a multi-mode interference (MMI) splitter, a directional coupler splitter, and the like. In some examples, the splitter **711** can split the input into any proportion such as 50% of the light along the light path **741** to DBR **713A** and 50% of the light along the light path **741** to DBR **713B**. The DBRs **713** can be configured as filters that allow certain wavelengths to propagate in its structure. The DBRs **713** can include a structure comprising multiple layers of materials having different refractive indices or different periodic characteristics. In some examples, the DBR **713A** may be configured as a different structure than the DBR **713B**, where the output to the detector **731A** can include a first set of wavelengths, and the output to detector **731B** can include a second set of wavelengths. For example, the DBR **713A** may include a stack of dielectric layers with different refractive indices than the stack of dielectric layers of the DBR **713B**.

(63) The detector **731A** and the detector **731B** may receive light from the DBR component **710A** and may generate signals indicative of the measured light. The signals may be transmitted along output **749A** and output **749B**. In some examples, the signals may be current signals. A controller (e.g., controller **112** illustrated in FIG. **1A**) may receive the signals, take the ratio of the signals, and

lock the monitored wavelength to the target wavelength by transmitting a signal to the light sources which may provide an adjustment to the light sources, for example, light sources **102** of FIG. **1A**. (64) In some instances, the DBR component may also output light along a light path, as shown in FIG. **7B**. Frequency stabilization subsystem **708B** may include a DBR component **710B**, which may be a selector that is configured to output a selected light path as the light path **742**. The light path **742** may be an output of the frequency stabilization subsystem **708B**. The DBR component **710B** can receive light on multiple light paths as inputs. For example, the light paths **144** of FIG. **1A** or the light paths **244** of FIG. **2** may be connected to the light paths **741** of FIG. **7**.

(65) The selector in the frequency stabilization subsystem may also include a filter component. FIG. **8A** illustrates a block diagram of an example frequency stabilization subsystem which may include a filter component and multiple detectors. Frequency stabilization subsystem **808A** can receive light on one or more light paths **841** as inputs and can generate one or more outputs, such as output **849A** and output **849B**. The frequency stabilization subsystem **808A** can include a filter component **810A** and detectors **831A** and **831B**. The detectors **831** can be correspondingly similar in function and structure as detectors **131** of FIG. **1A**.

(66) In some examples, the filter component **810A** can include an optical component **815** and a plurality of filters **821**. In some examples, the optical component **815** can include a waveguide and a DBR, which can output a first set of light to the filter **821A** and a second set of light to the filter **821B**. The filter **821A** can filter the light to output a first set of wavelengths to the detector **831A**, and the filter **821B** can filter the light to output a second set of wavelengths to the detector **831B**. The detector **831A** and the detector **831B** may receive light from the filter component **810A** and may generate signals indicative of the measured light. The signals may be transmitted along output **849A** and output **849B**. In some examples, the signals may be current signals. A controller (e.g., controller **112** illustrated in FIG. **1A**) may receive the signals, take the ratio of the signals, and lock the monitored wavelength to a target wavelength by transmitting a signal to the light sources which may provide an adjustment to the light sources, for example, light sources **102** of FIG. **1A**.

(67) In some instances, the filter component may also output light along a light path, as shown in FIG. **8B**. Frequency stabilization subsystem **808B** may include a filter component **810B**, which may be a selector that is configured to output light on a selected light path such as the light path **841**. The light path **841** may be an output of the frequency stabilization subsystem **808B**. The filter component **810B** can receive multiple light paths as inputs. For example, the light paths **144** of FIG. **1A** or the light paths **244** of FIG. **2** may be connected to the light paths **841**.

(68) As discussed herein, the illustrated block diagrams do not limit the configuration (e.g., location, size, orientation, etc.) of the optical routing lines and components. Examples of the disclosure can include other types of configurations not illustrated explicitly in the figures. Additionally or alternatively, the detector can be any type of detector, such as one that is discrete from the other components included in the frequency stabilization subsystem, or the detector may be an integrated component that is inseparable from the other components.

(69) Operation of the Optical Sensing System

(70) FIG. **9** illustrates a flow chart of an example operation of an optical sensing system having multiple detectors. At **952** of the process **950**, the controller can send one or more signals that control a plurality of light sources and at **954**, the plurality of light sources can receive the signal(s) and can emit light in response. The one or more signals from the controller can be indicative of one or more targeted properties of the light emitted by the plurality of light sources. At **956** of process **950**, a portion of the light emitted by the plurality of light sources can be directed using optics towards a system interface and at **958** another portion of the light (e.g., tapped light) emitted by the plurality of light sources can be directed towards a frequency stabilization subsystem.

(71) At step **960** of process **950**, one or more selectors included in the frequency stabilization subsystem can combine, multiplex, select, and/or filter at least a part of the tapped light and at **962**, the selector can output light to the detectors. The light output to the detectors can have one or more

unique properties such as different wavelengths, and so forth. At **964**, the detectors can receive the light and generate a signal indicative of the received light. The controller can receive and analyze the signals from the detectors at **966** and at **968** the controller can adjust the control signal(s) input to the plurality of light sources.

(72) Common Detector Arrangement

(73) A signal from a detector can drift over a given period of time (e.g., its lifetime). For example, the responsivity of a detector can drift. In instances where multiple detectors are used, the drifts of the multiple detectors may not be the same. The signals may include the respective drifts, which may not be detectable, and mathematical functions (e.g., taking the ratio) may not allow for accurate reduction or removal of these drifts from the signal(s).

(74) In some examples, a common detector can be used to remove or reduce the detector drift. FIG. **10A** illustrates a block diagram of an example frequency stabilization subsystem including an AWG and a common detector. Frequency stabilization subsystem **1008A** can include an AWG **1010A**, one or more attenuators **1037**, and a detector **1031**. The AWG **1010A** can have one or more functions and/or characteristics similar to AWG **510A** illustrated in FIG. **5A**. For example, the AWG **1010A** can include a plurality of waveguides **1015A**, where at least two waveguides can have different lengths. The AWG **1010A** can generate multiple outputs. A first output can be connected to the attenuator **1037A**, and a second output can be connected to the attenuator **1037B**. The attenuators **1037** can be variable optical attenuators (VOAs), for example. The attenuators **1037** can modulate the signals of the outputs from the AWG **1010A** at different frequencies.

(75) The output from one attenuator, such as the attenuator **1037A**, can be connected to one side of the waveguide of the detector **1031**, and the output from another attenuator, such as the attenuator **1037B**, can be connected to the other side of the waveguide of the detector **1031**. The detector **1031** can receive the attenuated signals and can generate a signal on the output **1049**. In this manner, the same detector **1031** can be used for multiple outputs from the AWG **1010A**, and the amount of drift from the detector **1031** included in the output **1049** can be reduced or removed. In some examples, the frequencies that the attenuators **1037** modulate the signals by can differ by 180°.

(76) Although FIG. **10A** illustrates one output **1049** from the frequency stabilization subsystem **1008A**, some examples may include using a selector that outputs light on a selected light path, such as AWG **510B** illustrated in FIG. **5B**.

(77) In examples where the multiplexer is a component separate from the selector, one light path can be a focal point of an adjacent grating order of the AWG **1010A**, and another light path can be multiplexed into the main grating order. For example, referring to FIG. **5B**, the light paths to the detectors **531A** and **B** can be the focal point of the adjacent grating order, and the light path **542** can be multiplexed into the main grating order. The properties (e.g., pitch and width) of the AWG **1010A** can be optimized such that some of the light along the light paths to the detectors **531A** and **B** can be scattered into the next grating order (e.g., the grating order adjacent to the main grating order).

(78) In some examples, the frequency stabilization subsystem may include other types of selectors and a common detector. For example, as shown in FIG. **10B**, frequency stabilization subsystem **1008B** can include a first ring resonator **1010B-1** and a second ring resonator **1010B-2** having one or more functionalities and/or characteristics similar to the first ring resonator **610A** and the second ring resonator **610B** illustrated in FIG. **6**. The ring resonators **1010B** can generate multiple outputs. A first output can be coupled (directly or indirectly) to the attenuator **1037A**, and a second output can be coupled to attenuator **1037B**. The attenuators **1037** can modulate the signals of the outputs from the ring resonators **1010B** at different frequencies. The output from one attenuator, such as attenuator **1037A**, can be coupled to one side of the waveguide of the detector **1031**, and the output from another attenuator, such as attenuator **1037B**, can be coupled to the other side of the waveguide of the detector **1031**. The detector **1031** can receive the attenuated signals and can generate a signal on the output **1049**.

(79) As another example, as shown in FIG. 10C, frequency stabilization subsystem **1008C** can include a DBR component **1010C** having one or more functionalities and/or characteristics similar to the DBR component **710A** illustrated in FIG. 7A or the DBR component **710B** illustrated in FIG. 7B. The DBR component **1010C** can include a splitter **1011** and multiple DBRs **1013A** and **1013B**. Similar to frequency stabilization subsystem **1008A** of FIG. 10A and frequency stabilization subsystem **1008B** of FIG. 10B, frequency stabilization subsystem **1008C** can be connected to one or more attenuators **1037** and a detector **1031**.

(80) As discussed herein, the illustrated block diagrams do not limit the configuration (e.g., location, size, orientation, etc.) of the optical routing lines and components. Examples of the embodiments described herein can include other types of configuration not illustrated explicitly in the figures. Additionally or alternatively, the detector can be any type of detector, such as one that is discrete from the other components included in the frequency stabilization subsystem, or the detector may be an integrated component that is inseparable from the other components.

(81) Detector Including Twin Waveguides

(82) In some examples, the drift in the detectors can be due to one or more material properties such as crystal defects. In some examples, the structure of the detectors may be changed to account for such defects. FIG. 11 illustrates a cross-sectional view of an example detector including multiple waveguides formed on the same substrate. The detector **1130** can include a substrate **1119**, where multiple waveguides, such as waveguide **1115A** and waveguide **1115B**, can be formed on the same substrate **1119**. In some examples, the waveguide **1115A** and the waveguide **1115B** can be twin waveguides, which can be waveguides, spaced with a small separation distance. The separation distance can be such that the wave of distortion from the crystal defects can affect both waveguides. For example, the separation distance **1198** may be in the approximate range of 2-5 μm and can be less than the waveguide's longitudinal dimension (e.g., 10 μm). The separation distance **1198** can be the distance between inner walls of the waveguides **1115A** and **1115B**.

(83) Multiple Light Sources

(84) Examples of the disclosure can further include the optical sensing system configured with a plurality of light sources. FIG. 12 illustrates a block diagram of an example optical sensing system including a frequency stabilization subsystem including an AWG and a detector.

(85) The optical sensing system can include a device **1200**. The device **1200** can include a plurality of light sources **1202**, a frequency stabilization subsystem **1208**, and a controller **1212**. The plurality of light sources **1202** can be configured to emit light along light paths **1244**. In some examples, the plurality of light sources **1202** can be configured to emit light having different ranges of wavelengths. Examples of the disclosure can include the plurality of light sources **1202** formed on multiple epitaxial chips.

(86) The frequency stabilization subsystem **1208** can include a selector, such as AWG **1210**, attenuators **1237**, and one or more detectors, such as detector **1231**. Although the figure illustrates an AWG **1210**, examples of the disclosure can include other selectors such as filters, resonators, multiplexers, MZIs, Fabry-Perot cavities, nanobeam cavities, ring resonators, DBRs, or the like. The selector (e.g., AWG **1210**), attenuators **1237**, and detector(s) **1231** can include one or more functionalities and/or characteristics similar to those disclosed herein. For example, the selector can have multiple functions: (1) multiplex, combine, filter, and/or select from its input(s) (e.g., light along the light paths **1244**), and (2) generate feedback information for controlling the light sources **1202** via, e.g., the controller **1212**. In some examples, FIG. 12 may not include the AWG **1210** and also may not include the attenuators **1237**. Continuing this example, FIG. 12 may include an MZI outputting light to two individual detectors **1231**.

(87) The frequency stabilization subsystem **1208** can output signals directly from the detector **1231** as output **1249** to the controller **1212**. In some examples, one or more components, such as filters, amplifiers, analog-to-digital converters (ADCs), etc. (not shown in FIG. 12) can receive signal(s) from the detector(s) **1231** and can output signal(s) to the controller **1212**.

(88) The signal(s) from the frequency stabilization subsystem **1208** can be used as feedback in, e.g., a control loop. A controller **1212** can receive and analyze the signal(s) from the frequency stabilization subsystem **1208** and can generate one or more signals. The signal(s) from the frequency stabilization subsystem **1208** can be the same signals used to control the light sources **1202** and the properties of light emitted by the light sources **1202**. In some examples, the signal(s) from the frequency stabilization subsystem **1208** (e.g., output **1249**) can be indicative of targeted changes in one or more properties (e.g., temperature, current, etc.) of the light sources **1202**. In some examples, the controller **1212** can include one or more other pieces of information (e.g., measured temperature of the light sources **1202**) in generating the signal(s).

(89) The frequency stabilization subsystem **1208** can also output light directed to the system interface for a measurement. In some examples, the selector can output the light to the system interface along the light path **1242**.

(90) Calibration and Thermal Stabilization

(91) The frequency stabilization subsystem can include one or more components for calibrating its temperature and can include one or more sensor components for detecting a change in its operation. The frequency stabilization subsystem can also include one or more heaters for controlling its temperature, which can affect its operation.

(92) FIG. **13A** illustrates a block diagram of an example ring resonator and calibration components. The ring resonator **1310B** can include a plurality of waveguides **1315B**. At least one waveguide, such as waveguides **1315B-2**, can be looped optical waveguides. One or more waveguides, such as waveguide **1315B-1**, can be an input waveguide, and one or more waveguides, such as waveguide **1315B-3** and waveguide **1315B-4**, can be output waveguides. Input waveguides can be waveguides connected to light input to the ring resonator **1310B**, and output waveguides can be waveguides connected to light output from the ring resonator **1310B**. In some instances, the input and output waveguides may include the same structure. A looped waveguide **1315B-2** can include a closed loop that is optically coupled to the input waveguide **1315B-1** and one of the output waveguides **1315B-3** or **1315B-4**.

(93) The ring resonator **1310B** can include calibration components such as heaters **1323** and sensors **1325**. The sensors **1325** can be temperature sensors, for example, that measure the temperature of the ring resonator **1310B**. If the temperature of the ring resonator **1310B** deviates within a certain threshold(s) from the targeted temperature (e.g., does not meet a temperature criteria), a controller (not shown) can send one or more control signals to the heaters **1323** to change the temperature of the ring resonator **1310B** to within the targeted threshold(s).

(94) The calibration components can be included in other types of components, such as a DBR, as shown in FIG. **13B**. DBR **1313** can include calibration components such as heater **1323** and sensor **1325**. Similar to the calibration components discussed above, the sensor **1325** can measure the temperature of the DBR **1313**, and this temperature measurement can be used as feedback to the heater **1323** for controlling (e.g., adjustment) the temperature of the DBR **1313**. Although a single heater **1323** and a single sensor **1325** are illustrated in FIG. **13B**, examples can include any number of heaters, any number of sensors, and any ratio of heaters to sensors. In some examples, the number of heaters and the number of sensors can depend on the size of the component. For example, larger passive components can have a greater number of calibration components than smaller passive components.

(95) Examples of the disclosure can include executing the calibration procedure at periodic intervals, upon system startup, in synchronization with the measurements, when the light emitted by the light sources deviates from its targeted properties, and so forth. Although FIGS. **13A-13B** illustrate the calibration components as included in a ring resonator and DBR, respectively, examples of the disclosure can include calibration components as included in other types of passive components, such as those described herein. Additionally, examples can include the heaters and sensors having configurations (e.g., location, size, orientation, etc.) not explicitly shown in the

figures. For example, heater 1323 in FIG. 13A may be square or any other appropriate shape, instead of rectangular.

(96) Although the disclosed examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosed examples as defined by the appended claims.

(97) The foregoing description, for purposes of explanation, uses specific nomenclature to provide a thorough understanding of the described embodiments. However, it will be apparent to one skilled in the art, after reading this description, that the specific details are not required in order to practice the described embodiments. Thus, the foregoing descriptions of the specific embodiments described herein are presented for purposes of illustration and description. They are not targeted to be exhaustive or to limit the embodiments to the precise forms disclosed. It will be apparent to one of ordinary skill in the art, after reading this description, that many modifications and variations are possible in view of the above teachings.

(98) The present disclosure recognizes that personal information data, including the biometric data acquired using the presently described technology, can be used to the benefit of users. For example, the use of biometric authentication data can be used for convenient access to device features without the use of passwords. In other examples, user biometric data is collected for providing users with feedback about their health or fitness levels. Further, other uses for personal information data, including biometric data that benefit the user are also contemplated by the present disclosure.

(99) The present disclosure further contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure, including the use of data encryption and security methods that meets or exceeds industry or government standards. For example, personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection should occur only after receiving the informed consent of the users. Additionally, such entities would take any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices.

(100) Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data, including biometric data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, in the case of biometric authentication methods, the present technology can be configured to allow users to optionally bypass biometric authentication steps by providing secure information such as passwords, personal identification numbers (PINs), touch gestures, or other authentication methods, alone or in combination, known to those of skill in the art. In another example, users can select to remove, disable, or restrict access to certain health-related applications collecting users' personal health or fitness data.

Claims

1. A system comprising: a light source that emits light along an input light path; a selector comprising an arrayed waveguide grating configured to: receive light along the input light path; output light of a first wavelength along a first output of the arrayed waveguide grating; output light

of a second wavelength along a second output of the arrayed waveguide grating; and output light along a third output of the arrayed waveguide grating; a first detector that receives the output light of the first wavelength from the first output; a second detector that receives the output light of the second wavelength from the second output; and a controller that: receives signals generated by the first detector and the second detector; determines a ratio of the signals generated by the first detector and the second detector; and generate a control signal for the light source based on the determined ratio, wherein the control signal controls the light source to lock a wavelength of light emitted by the light source along the third output to a target wavelength.

2. The system of claim 1, comprising: a plurality of light sources that includes the light source that emit light along a plurality of light paths that includes the input path, wherein the arrayed waveguide grating receives light from the plurality of light sources along the plurality of light paths.

3. The system of claim 1, comprising: a plurality of light sources that includes the light source that emit light along a plurality of light paths; and a multiplexer that combines light from the plurality of light paths to a common light path.

4. A method, comprising: emitting light from a light source along an input light path; receiving, by a selector comprising an arrayed waveguide grating, light along the input light path; outputting, from the selector, light of a first wavelength along a first output of the arrayed waveguide grating; outputting, from the selector, light of a second wavelength along a second output of the arrayed waveguide grating; outputting, from the selector, light along a third output of the arrayed waveguide grating; receiving, at a first detector, the light of the first wavelength receiving, at a second detector, the light of the second wavelength; outputting signals from the first detector and the second detector; determine a ratio between the signals; and controlling the light source based on the determined ratio to lock a wavelength of the light outputted along the third output to a target wavelength.

5. A system comprising: a light source that emits light along an input path; an arrayed waveguide grating configured to: receive light along the input light path; output light of a first range of wavelengths along a first output; output light of a second range of wavelengths along a second output; output light along a third output; a first detector that receives the output light of the first range of wavelengths from the first output of the arrayed waveguide grating; a second detector that receives the output light of the second range of wavelengths from the second output of the arrayed waveguide grating; and a controller that: receives signals generated by the first detector and the second detector; determines a ratio of the signals generated by the first detector and the second detector; and generate a control signal for the light source based on the determined ratio, wherein the control signal controls the light source to lock a wavelength of light emitted by the light source along the third output of the arrayed waveguide grating to a target wavelength.

6. The system of claim 5, comprising: a plurality of light sources that includes the light source that emit light along a plurality of light paths that includes the input path, wherein the arrayed waveguide grating receives light from the plurality of light sources along the plurality of light paths.

7. The system of claim 5, comprising: a plurality of light sources that includes the light source that emit light along a plurality of light paths; and a multiplexer that combines light from the plurality of light paths to a common light path.
